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line Catalog

Stainless Steel 17-4PH

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17-4PH is a widely-used martensitic precipitation-hardening stainless steel with a high concentration of copper, nickel, and niobium. It has many favorable properties, including excellent strength, good corrosion resistance, and excellent mechanical properties up to 600 °F (316 °C).

17-4PH also possesses good toughness when welded, and short low-temperature heat treatments reduce aging and scaling. Additionally, 17-4PH is easy to machine and fabricate.

Due to the beneficial properties, 17-4PH finds use in many industries. Typical parts include mechanical seals, oil patch, and pump shafts.

Furthermore, the alloy also finds use in the aerospace industry, petroleum industry, and chemical industry, food processing equipment, paper and pulp industry, nuclear waste processing and storage, and general engineering working.

[Steel & Materials](#) offers Stainless Steel 17-4PH in two sub-type specifications and multiple shapes/forms:

[AMS 5604](#) (Plate, Sheet, and Strip)

[AMS 5643](#) (Bar, Forging, Ring, Tubing, and Wire)

Stainless Steel 17-4PH Chemical Composition

Element	Min (%)	Max (%)
Carbon, C	—	0.07
Manganese, Mn	—	1
Niobium, Nb	—	1
Nickel, Ni	—	0.03
Phosphorus, P	—	0.04
Chromium, Cr	15	17.5

Request a Quote

Name *

Company *

Email *

Phone *

Alloy spec *

17-4PH

Shape *

Size/Diameter *

Measure *

Inches

Length *

Measure *

Inches

Qty *

Comments

ium + Tantalum, Nb + Ta	0.15	0.45
al, Ni	3.0	5.0
er, Cu	3.0	5.0
Fe	—	*Balance

clusively to the element mentioned, but that one predominates other elements that are used in smaller ties.

rication and Working Instructions

ess Steel 17-4PH reacts best to hot forming at 1742 to 2192 °F (950 to 1200 °C), with annealing at 76 °F (°) and aging. The alloy can also be cold-formed in limited form, primarily to plates in the annealed ion. However, stress corrosion resistance can be improved with re-aging.

Moreover, 17-4PH can be cut using mechanical operations, such as abrasive waterjets, machining, aw, and shearing. However, plasma cutting isn't recommended on this alloy.

H has the best welding characteristics of competing stainless steel alloys. Moreover, it can be welded all standard techniques, including gas tungsten arc, gas metal arc, shielded metal arc, and plasma arc. To re desired mechanical properties, pre-weld and post-weld heat treatments are recommended. For the best s, we recommend the AMS 5643 wire.

sical Properties of Stainless Steel 17-4PH

erty	Imperial	Metric
ng Range	2560-2625 °F	1404-1440 °C
ific Heat	0.11 Btu/lb.-°F	460 Joules/gg-K
ific Density	0.282 lb/in3	7.8 g/cm3
rical Resistivity	38.6 μΩ in	98 μΩ cm
ar Coefficient of Thermal Expansion (70 °F to 200 °F) (21 °C to 2)	6 [in/in/°F·106]	10.8 [μm/m·°C]
ar Coefficient of Thermal Expansion (70 °F to 400 °F) (21 °C to 206)	6 [in/in/°F·106]	10.8 [μm/m·°C]
ar Coefficient of Thermal Expansion (70 °F to 600 °F) (27 °C to 316)	6.2 [in/in/°F·106]	11.2 [μm/m·°C]
ar Coefficient of Thermal Expansion (70 °F to 800 °F) (21 °C to 427)	6.3 [in/in/°F·106]	11.2 [μm/m·°C]
mal Conductivity (@ 300 °F) (@ 149 °C)	124 Btu/(hr/ft²/in/°F)	17.9 [W/m-K]
mal Conductivity (@ 500 °F) (@ 260 °C)	135 Btu/(hr/ft²/in/°F)	19.5 [W/m-K]
mal Conductivity (@ 860 °F) (@ 460 °C)	156 Btu/(hr/ft²/in/°F)	22.5 [W/m-K]
mal Conductivity (@ 900 °F) (@ 482 °C)	157 Btu/(hr/ft²/in/°F)	22.6 [W/m-K]
son's Ratio (H900 Condition)	0.272	0.272
ulus of Elasticity (H900 Condition)	28 x 10 ⁶ ksi	197 x 10 ³ MPa
ulus of Rigidity in Torsion	9.68 x 10 ³ ksi	67 x 10 ³ MPa

nless Steel 17-4PH Mechanical Properties

erty	Imperial	Metric
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For more information, call: 631-685-

Metal Weight Calculator

Make use of our online metal weight calculator to calculate the weight of Stainless Steel 17-4PH

Material

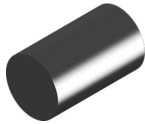
Stainless Steel

Alloy

17-4PH

Shape

Bar



Diameter

Inches

Length

Inches

Quantity

1

Weight

0 Lbs

CALCULATE

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Metal Weight Calculator - [Disclaimer](#)

ate Tensile Strength	23,300 psi	160 MPa
Strength (0.02%)	16,680 psi	115 MPa
gation (% in 2")	5%	5%
well Hardness C	35	35

Heat Treatment

304 Stainless Steel 17-4PH is available in the solution annealed condition at 1900 °F (1038 °C) and then air-cooled to 100 °F (32 °C).

However, additional heat treatments can produce various hardness and toughness levels. The most common heat treatments are listed in the table below.

Condition	Temperature [± 15 °F (± 8.4 °C)]	Cooling Method and Duration
0	900 °F (482 °C)	Air cooling for 1 hour
5	925 °F (496 °C)	Air cooling for 4 hours
25	1025 °F (551 °C)	Air cooling for 4 hours
75	1075 °F (580 °C)	Air cooling for 4 hours
100	1100 °F (593 °C)	Air cooling for 4 hours
50	1150 °F (621 °C)	Air cooling for 4 hours
50 + 1150	1150 °F (621 °C) followed by 1150 °F (621 °C)	Air cooling for 4 hours followed by Air cooling for 4 hours
50-M	1400 °F (760 °C) followed by 1150 °F (621 °C)	Air cooling for 2 hours followed by Air cooling for 4 hours

Select AMS Number:

AMS Number	Alloy	Type	UNS	Cross Ref. Spec	Misc./Shape	
AMS 5604	17-4PH	Stainless Steel	S17400			
AMS 5604 Plate	17-4PH	Stainless Steel	S17400	-	Plate	
AMS 5604 Sheet	17-4PH	Stainless Steel	S17400	-	Sheet	
AMS 5604 Strip	17-4PH	Stainless Steel	S17400	-	Strip	
AMS 5643	17-4PH	Stainless Steel	S17400			
AMS 5643 Bar	17-4PH	Stainless Steel	S17400	-	Bar	
AMS 5643 Forging	17-4PH	Stainless Steel	S17400	-	Forging	
AMS 5643 Ring	17-4PH	Stainless Steel	S17400	-	Ring	
AMS 5643 Tubing	17-4PH	Stainless Steel	S17400	-	Tubing	

Number	Alloy	Type	UNS	Cross Ref. Spec	Misc./Shape
5643 Wire	17-4PH	Stainless Steel	S17400	-	Wire 



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 631-218-0013
 631-218-0141

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	Tungsten

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